

 **DARPA TECHNICAL & COST PROPOSAL**

PROJECT HYPERION: Zero-Trust Adaptive Mesh Neural Routing for Contested Electromagnetic Environments

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BAA Number:	BAA-DARPA-I2O-2026-09	Submitting Org:	Apex Defense Systems, Inc.
Technical Topic:	Resilient Tactical Comms	Team Partners:	Vertex Labs, Meridian Micro
Principal Investigator:	Dr. Marcus Vance	Period of Perf.:	24 Months (Base + Option)
PI Contact Email:	m.vance@apexdef.com	Total Proposed Cost:	\$3,845,200 (CPFF Contract)

TOTAL BUDGET
\$3,845,200

PERFORMANCE PERIOD
24 Months

TARGET MATURITY
TRL 3 → TRL 6

SECURITY CLEARANCE
TOP SECRET / SCI

1. EXECUTIVE SUMMARY & PROPOSAL OVERVIEW

Modern tactical operations encounter unprecedented electromagnetic spectrum (EMS) warfare, characterized by high-powered directional jamming, cognitive RF spoofing, and instantaneous node compromise. Legacy military ad-hoc networks (MANETs) rely on centralized route discovery and predictable frequency hopping schedules, leaving tactical units vulnerable to total communication blackouts within seconds of adversarial engagement. **Project HYPERION** directly addresses this critical vulnerability by introducing a decentralized, zero-trust neuromorphic routing architecture. Designed for deployment at the tactical edge, HYPERION leverages embedded low-power tensor processing units (TPUs) to dynamically synthesize interference-free communication channels in real time without relying on GPS, fixed infrastructure, or centralized command nodes.

 **Key Innovation 1: Edge Neural Spectrum Synthesis**

- Real-time RF spectrum sensing via deep reinforcement learning running at sub-50µs inference latency.
- Predicts adversarial jamming sweeps prior to channel degradation, initiating preemptive waveform synthesis.
- Zero dependency on satellite timing references or pre-shared hopping tables.

 **Key Innovation 2: Cryptographic Zero-Trust Mesh**

- Micro-second zero-knowledge proof (ZKP) identity verification for every transmitted data packet.
- Instantaneous detection and isolation of compromised or spoofed friendly nodes within 1.2 milliseconds.
- Quantum-resistant lattice cryptography optimized for SWaP-constrained field radios.

1.1 Statement of Work & High-Level Objectives

Apex Defense Systems, in partnership with Vertex Research Laboratories and Meridian Microelectronics, proposes a 24-month effort structured in two distinct phases:

- Objective 1: Phase I (Base – 12 Months):** Formulate and validate the HYPERION Neuromorphic Algorithm Suite, construct a high-fidelity Hardware-in-the-Loop (HITL) RF simulation rig, and demonstrate TRL 4 capability across a 20-node simulated mesh under heavy adversarial jamming.
- Objective 2: Phase II (Option – 12 Months):** Fabricate custom software-defined radio (SDR) mezzanine cards with Meridian TPU accelerators, integrate onto tactical unmanned aerial vehicles (UAVs) and ground units, and execute a live-fire field test achieving TRL 6 validation.